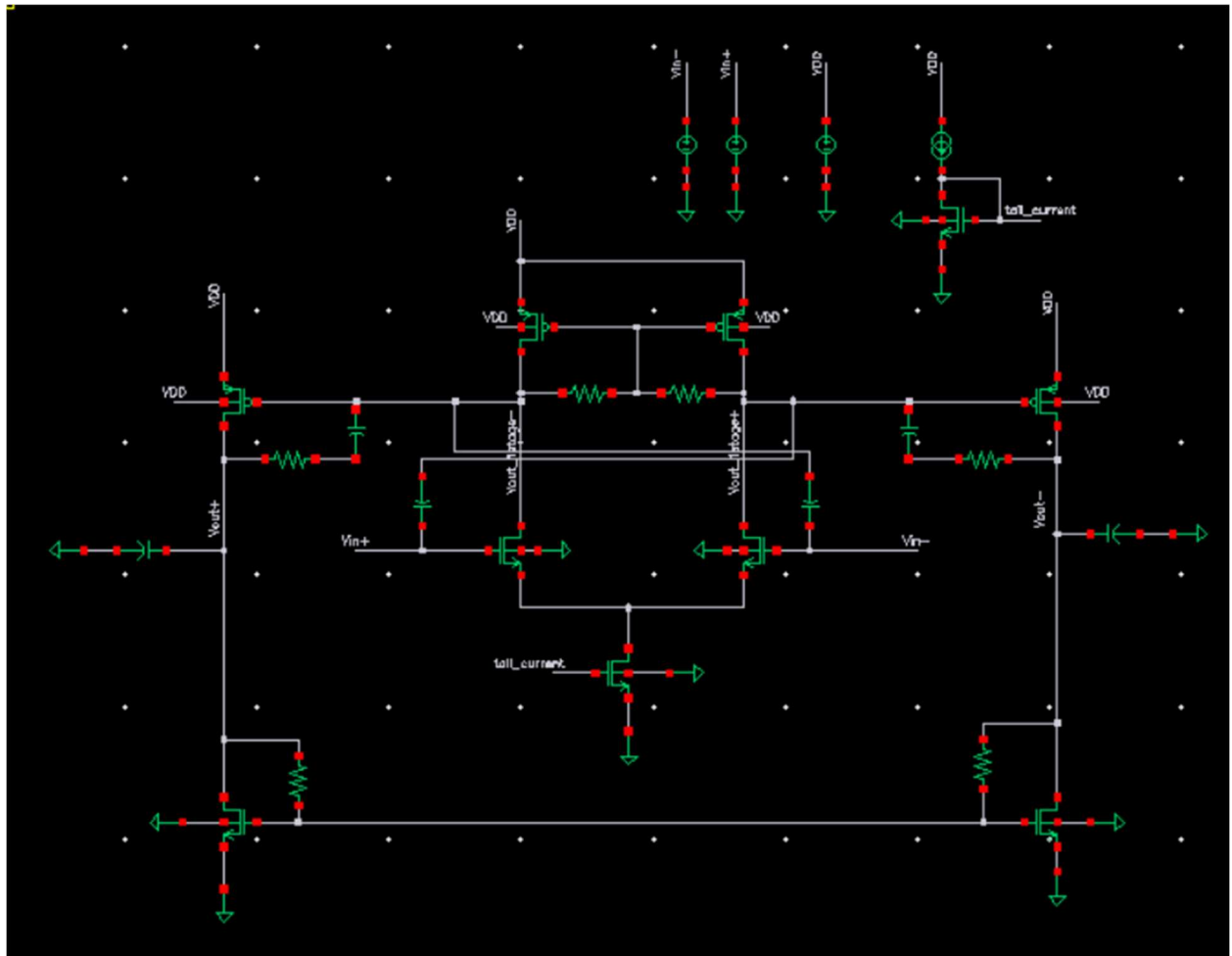


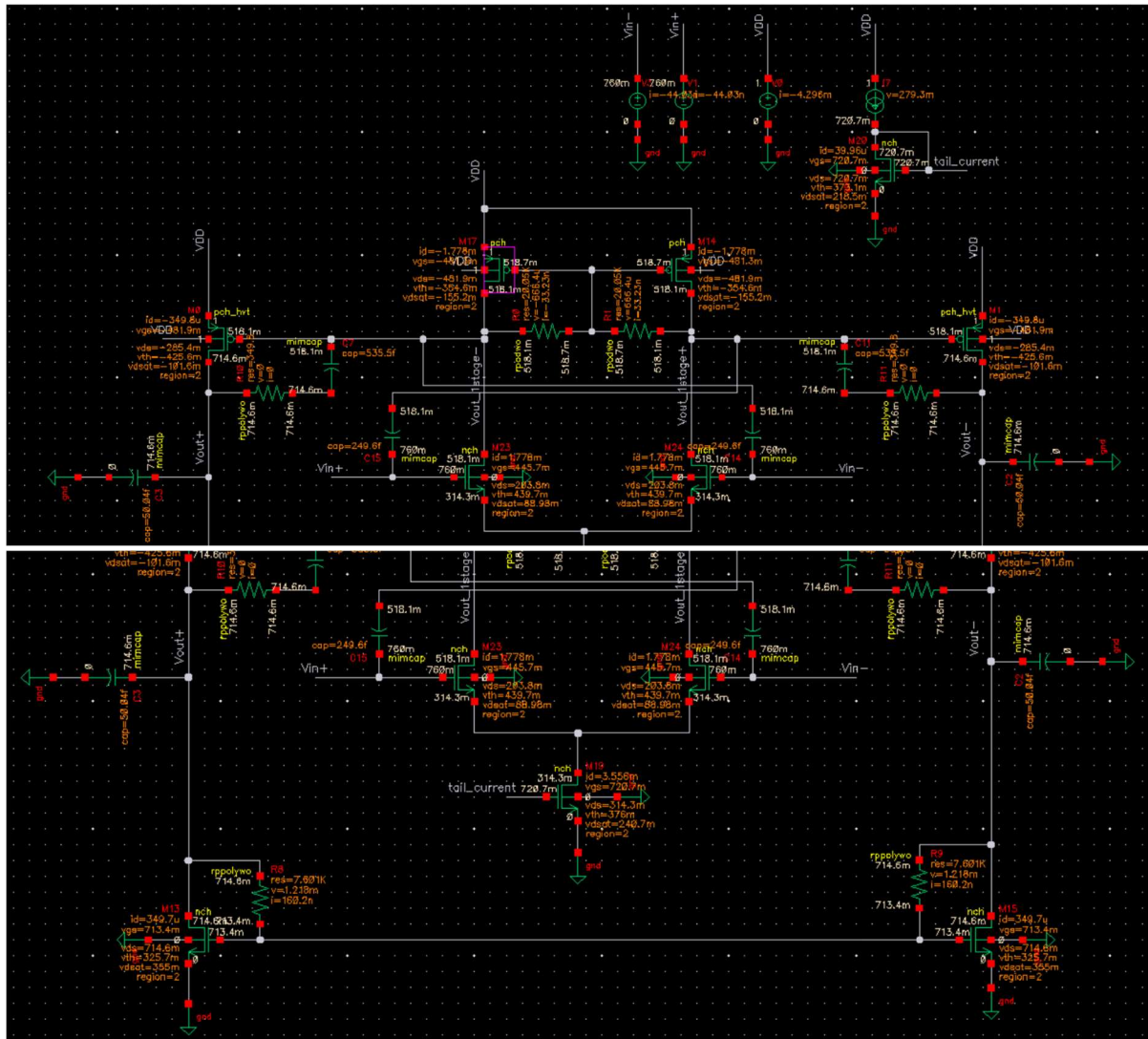
Specifications

	Designed Circuit's Spec	Target
Gain	41.2 dB	> 40dB
Unity Gain Frequency	5.05 GHz	> 5 GHz
Phase Margin	61.4 degrees	> 60 degrees
Power Consumption	4.3 W	< 4.5 W
Pk-Pk Amplitude (Single-Ended)	543.4 mV	> 400mV

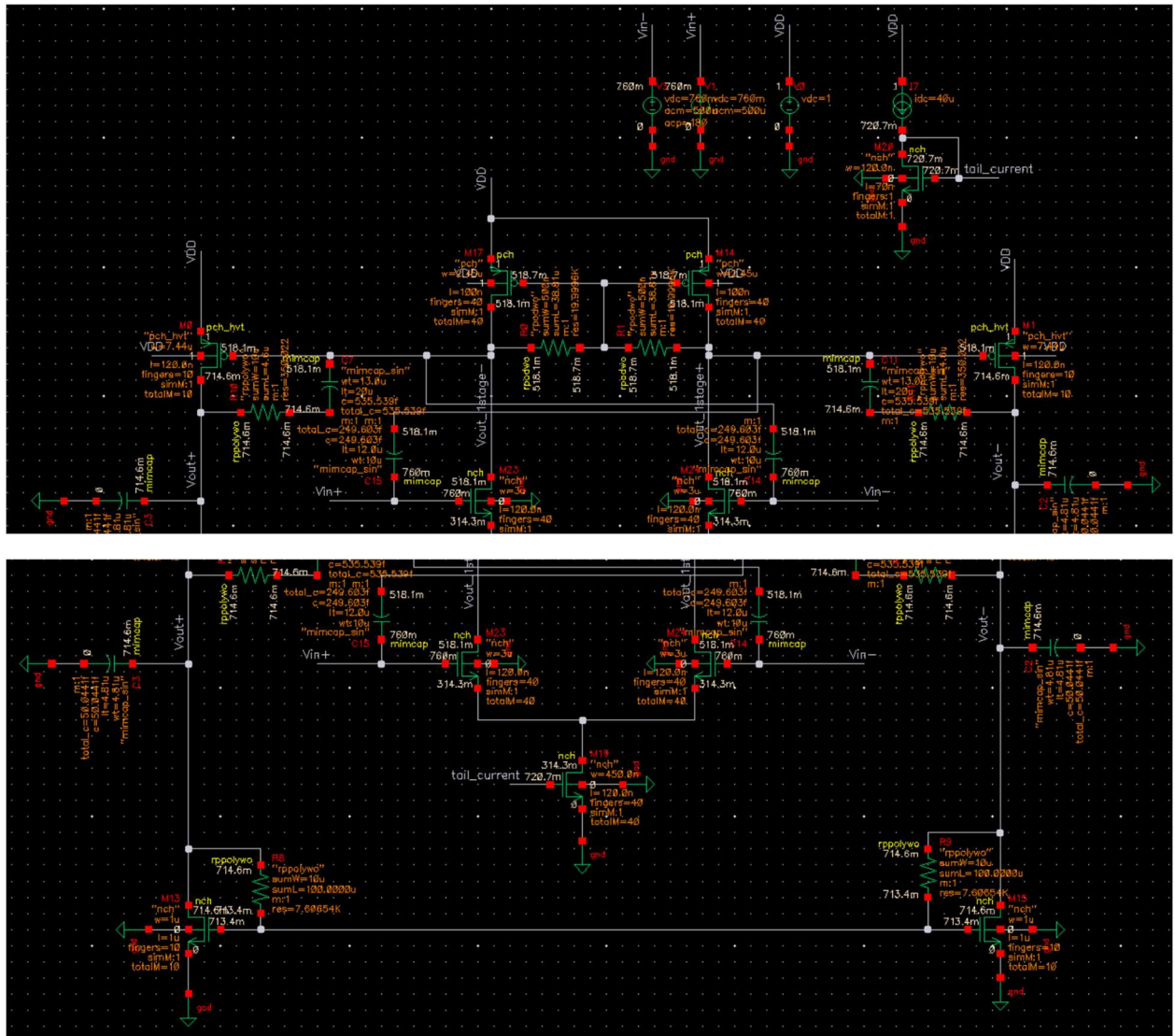
Overall Design



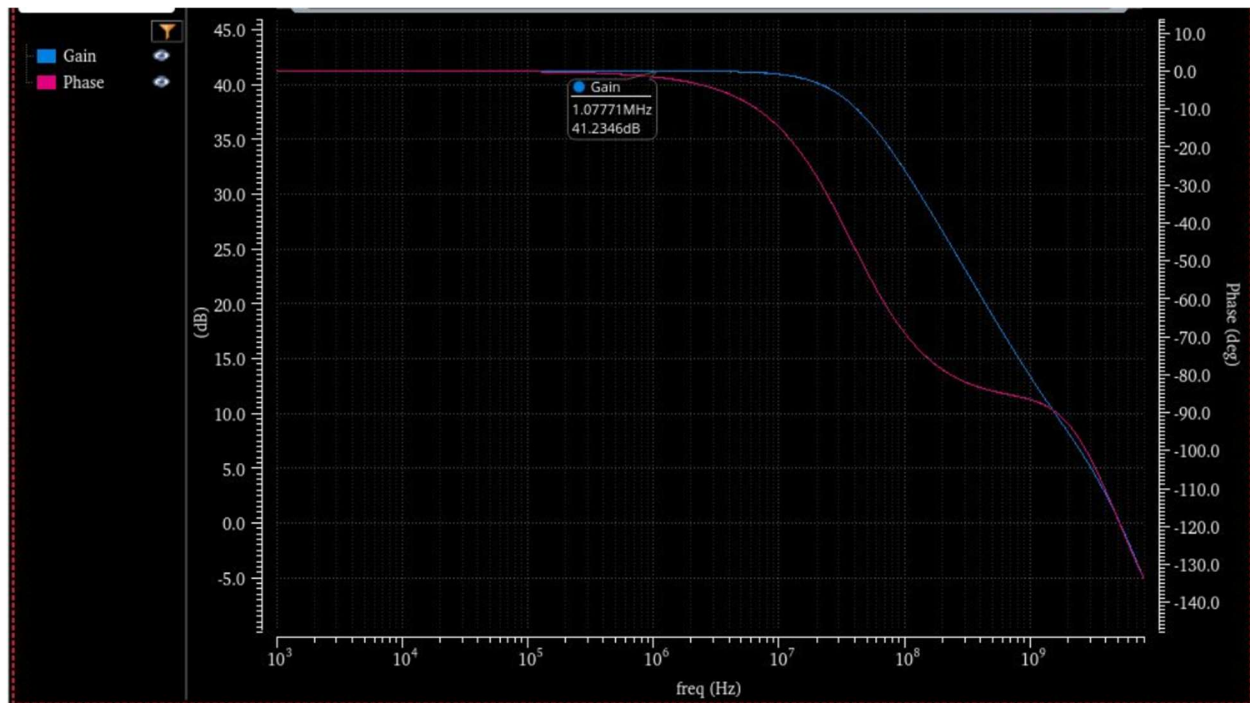
DC Operating Points & DC Node Voltages



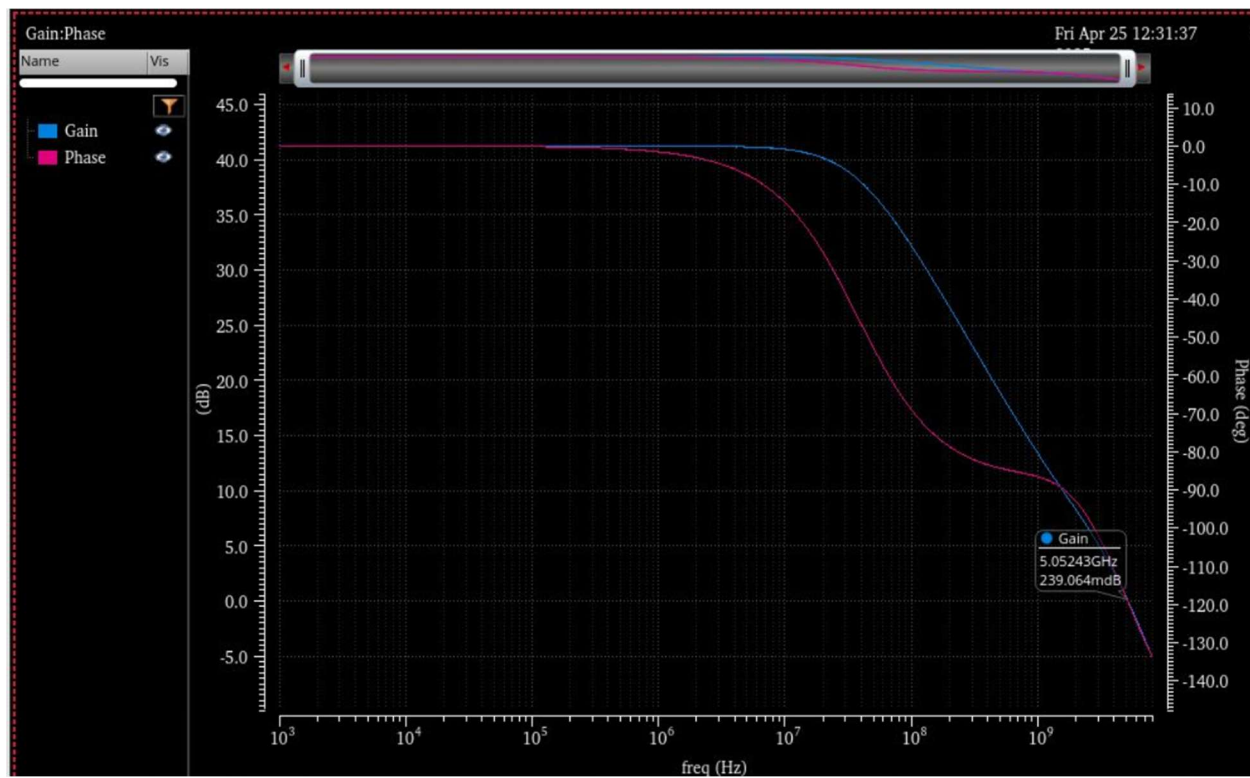
Component Parameters



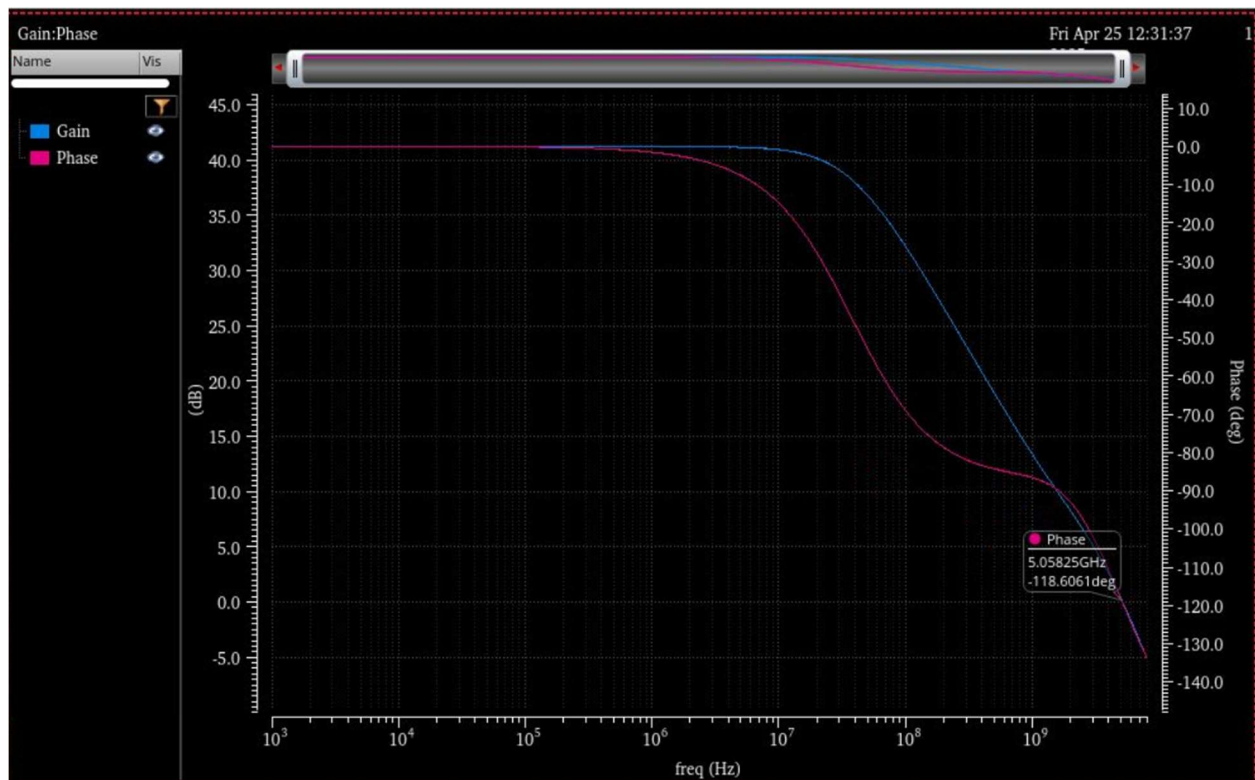
Gain = 41.23 dB



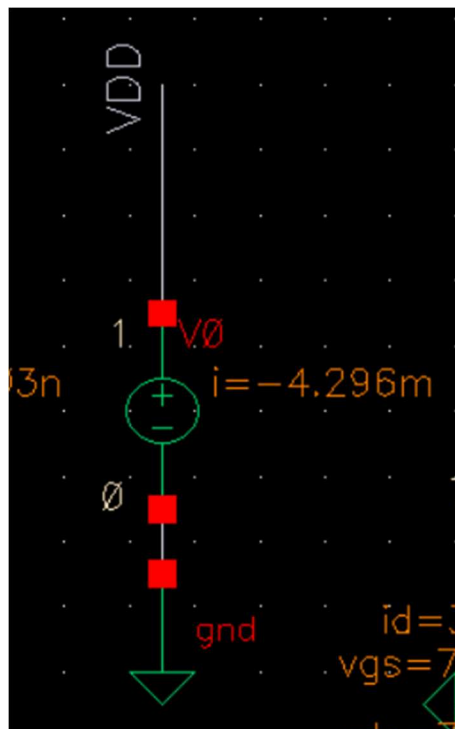
Unity Frequency = 5.1 GHz



Phase Margin = 51.4 degrees



Power Consumption = 4.296W



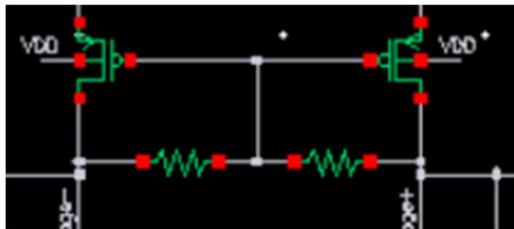
Peak-to-Peak Voltage at the Output Node (Single Ended)

$$= V_{dd} - V_{d,sat} \text{ of } M13 - V_{d,sat} \text{ of } M0$$

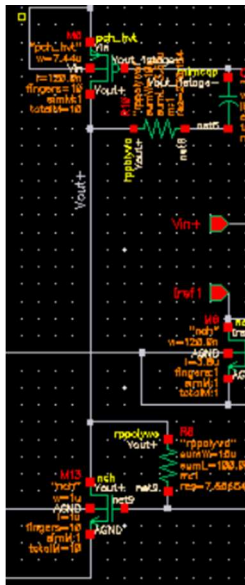
$$= 1 \text{ V} - 355 \text{ mV} - 101.6 \text{ mV}$$

$$= 543.4 \text{ mV}$$

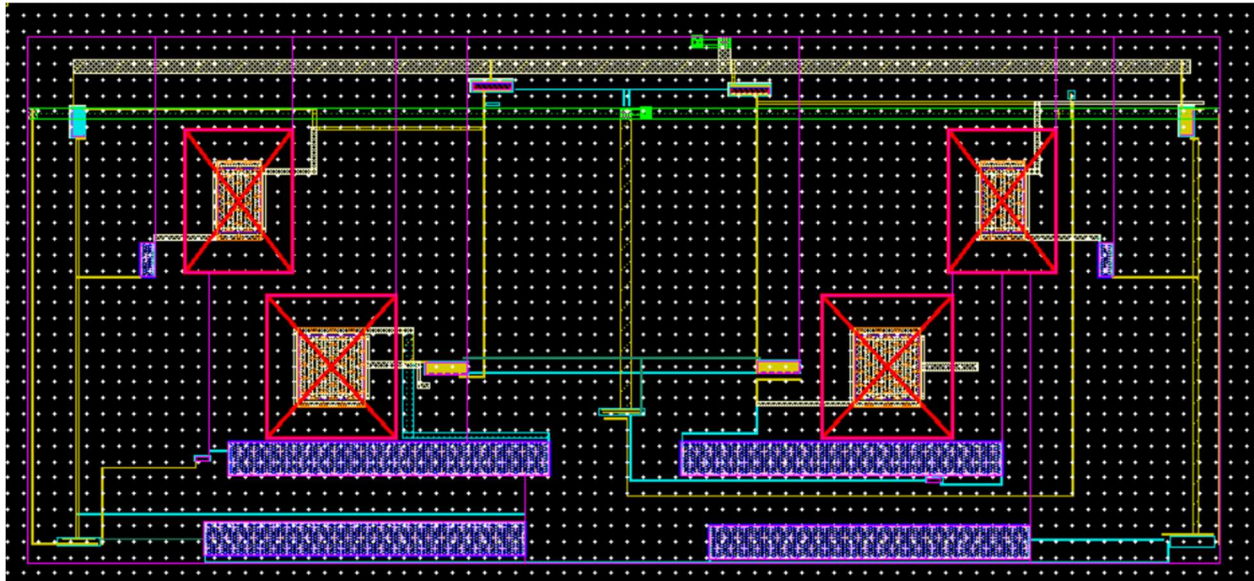
Common-Mode Feedback Used



Feedback Loops Used



Layout



DRC & LVS Clean

Cell/Rule	Color	Count	Level
TOP VLSI_Midterm_layout		15	0
DRM.R.1		1	
M1.DN.1		1	
M2.DN.1		1	
M3.DN.1		1	
M4.DN.1		1	
M5.DN.1		1	
M6.DN.1		1	
M7.DN.1		1	
M8.DN.1		1	
M8.DN.5:L		1	
M9.DN.2		1	
OD.DN.2		1	
PO.DN.2		3	

